

# Customer Segmentation using Clustering

## K-Means • Mall Customers Dataset • Week 4 Major Project

### 1. Introduction

Customer segmentation divides customers into groups with similar characteristics or behavior. This project applies K-Means clustering to age, annual income and spending score.

### 2. Problem Statement

The objective is to identify customer groups without predefined class labels and present the groups through an interactive business analytics dashboard.

### 3. Dataset

The Week 4 brief specifies the Kaggle Mall Customers Dataset. The expected fields are CustomerID, Gender, Age, Annual Income (k\$), and Spending Score (1-100).

### 4. Workflow

Data cleaning → feature selection → StandardScaler → elbow method and silhouette score → K-Means → cluster profiling → spending-based labels → visualization → business interpretation.

### 5. K-Means

K-Means is an unsupervised learning method that partitions observations into K clusters around learned centroids. The assignment version uses K=3 so the resulting clusters can be described as Low Spenders, Medium Spenders and High Spenders.

### 6. Segment Labels

Clusters are ordered by their mean Spending Score. The lowest mean is labelled Low Spenders, the middle mean Medium Spenders, and the highest mean High Spenders.

### 7. Visualizations

The application includes age distribution, income-versus-spending, age-versus-spending, elbow curve, silhouette curve, interactive 3D visualization and PCA visualization.

### 8. Business Interpretation

Low Spenders can be studied for retention and product discovery. Medium Spenders can be considered for cross-selling and bundles. High Spenders can be considered for loyalty and premium engagement. These are descriptive business interpretations, not guaranteed future behavior.

### 9. Limitations

Results depend on feature choice, scaling, K and the input dataset. K-Means requires a chosen number of clusters, and cluster labels are descriptive rather than predictive.

### 10. Future Scope

Add transaction history, purchase frequency and transaction value; compare clustering algorithms; connect the system to a database; add recommendation functionality; and deploy the Streamlit dashboard.

## **11. Conclusion**

The project demonstrates a complete unsupervised machine learning pipeline and converts clustering results into an interactive customer analytics application.